Chemistry: Grade 2, Semester 1

Change, Equilibrium and Cycles

Big Idea: Water is a universal solvent. Aqueous solutions can have different chemical and physical properties which are affected by different factors. Quantitative and qualitative analysis are tools to study solutions.

CH.2.01 - Outline the purification of the water supply.

Week 01 - Week 02

Learning Outcome: Outline the purification of the water supply. Describe two different processes whereby sea water can be converted into drinkable water.

Key Concepts:

1. Solutions and aqueous solutions.
2. Solubility of salts (temperature impact).
3. Solubility curves.
4. Quantitative analysis.
    - ppm.
    - molarity.
    - molality.
    - normality.
    - mole fraction.
    - mass percentage.
    - TDS.
5. The different ion percentages which the human body needs daily.
6. Resins.
7. EDTA (including ion exchange).
8. Electrodialysis.
9. Reverse osmosis.
10. Dissolved oxygen (DO).
11. Qualitative analysis.

Skills: 1. Use simple distillation to separate soluble salt 2. Use filtration to separate in soluble salt 3. Use carbon to remove tastes and colors 4. Calculate rates of chlorination to disinfect the water 5. Compare between chemical, biological and physical treatment 6. Determine the concentration of solutes by different methods 7. Compare between

qualitative and qualitative analysis. 8. Use analytically equipment correctly 9. interpret the analytical data and results.
Essential Questions: How can you get a pure drinkable water?

Textbook and Resource Materials: How can you get a pure drinkable water?

Evidence of Learning: 1. Use litmus paper for determination of pH 2. Determine pH by use of meter 3. Experimentally demonstrate multiple methods of analysis of pH

Capstone Connection: Because water is such a good solvent it creates special challenges in terms of pollution and water supply.

Relations: MA.2.01; () BI.2.02; () ES.2.04; ()

SEC Topic & Code: HSST-CH 2.05 (HSST-CH 2.05)

CH.2.02 - Demonstrate that water is a good solvent.

Week 03 - Week 04
Learning Outcome: Demonstrate that water is a good solvent, that it contains different varieties of dissolved substances, and identify environmental sources and the effect on water purity.

Key Concepts:

1. Water in the environment.
2. Properties and types.
3. Measuring densities, conductivity, pH, boiling points.
4. Effect of solutes on physical properties.
5. Colligative properties.
    a. elevation in B.P.
    b. depression in F.P.
    c. osmotic pressure.

Skills: 1. Identify naturally occurring dissolved and suspended materials in water sources (mineral salts; oxygen; organic matter) 2. Describe pollutants, their origins and the chemical interactions that make them harmful (metal compounds; sewage; nitrates from fertilizers; phosphates from fertilizers and detergents; harmful microbes) 3.

Develop chemical explanation for the role of beneficial chemicals in water (e.g. oxygen and mineral salts for aquatic life)

Essential Questions: Why do we add ethylene glycol to water in car's radiator?

Textbook and Resource Materials: Why do we add ethylene glycol to water in car's radiator?

Evidence of Learning: 1. Experimentally analyze physical and chemical properties of water from different sources.

Capstone Connection: Impact of dissolved substances in the treatment of water

Relations: MA. 2.07; ()

SEC Topic & Code: HSST-CH 2.04 HSST-CH 2.05 (HSST-CH 2.04 HSST-CH 2.05)

CH.2.03 - Describe characteristic properties of acids and bases.

Week 05 - Week 06
Learning Outcome: Describe characteristic properties of acids and bases and preparation of salts

Key Concepts:

1. acids
    a. nature of acids.
    b. acid strength.
    c. pH scale.
2. Polyprotic acid.
3. Bases.
4. Acid-base properties of salts.
5. The Effect of Structure on.
    a. acid–base properties.
    b. acid–base properties of oxides.
6. The Lewis Acid–Base Model.

Skills: 1. Describe the techniques used in the preparation, separation and purification of salts as examples of some of the techniques specified in the previous LO (methods for

preparation should include precipitation and titration together with reactions of acids with metals, insoluble bases and insoluble carbonates) 2. Describe the general rules of solubility for common salts to include nitrates, chlorides (including silver and lead), sulfates (including barium, calcium and lead), carbonates, hydroxides, Group I cations and ammonium salts 3. Suggest a method of preparing a given salt from suitable starting materials, given appropriate information 4. Solving acid- base problems

Essential Questions: How do acids, bases and salts affect our lives?

Textbook and Resource Materials: How do acids, bases and salts affect our lives?

Evidence of Learning: 1. Experimentally analyze titrations of mono-, di- and tri-protic acids 2. Describe the ionization of strong acids in water and the dissociation of strong bases in water. 3. Distinguish between solutions of weak acids or bases and solutions of strong acids or bases. 4. Relate pH to the concentration of hydronium ions and hydroxide ions in a solution. - Write ionic equations for neutralization reactions. 5. Identify the products of a neutralization reaction. 6. Describe the composition of a salt."

Capstone Connection: Role of acids and bases in water treatment or measurement

Relations: ES.2.13; ()

SEC Topic & Code: HSST-CH 2.04 (HSST-CH 2.04)

CH.2.04 - Describe the effect of concentration, pressure, surface area, temperature.

Week 07 - Week 08

Learning Outcome: Describe the effect of concentration, pressure, surface area, temperature and catalysis (including enzymes) on the rates of reactions and explain these effects in terms of collisions between reacting particles.

Key Concepts:

1. Order.
2. Molecularity.
3. Rate of chemical reaction.
4. Law of mass action.
5. Determination of rate of reaction.

Skills: 1. Determine the rate of chemical reaction by monitoring: a. change of concentration of reactant or product. b. volume of gas evolved c. change in PH d. change of weight of the reactants.

Essential Questions: why the food spoils outside the fridge in hot days?

Textbook and Resource Materials: why the food spoils outside the fridge in hot days?

Evidence of Learning: Sample Exercise Zumdahl, 12.2 - Students must understand how the study of the rates of chemical reactions and the structures of the products of these reactions can lead to knowledge of the detailed atomic-level behavior of chemical substances and elucidation of their chemical and physical properties.

Capstone Connection: Effect of these parameters on water treatment for potable water applications

Relations: MA.2.05; ()

SEC Topic & Code: HSST-CH 2.05 (HSST-CH 2.05)

CH.2.05 - Explain how catalysts (transition metals, nanoparticles, and biological).

Week 09 - Week 10

Learning Outcome: Explain how catalysts (transition metals, nanoparticles, and biological enzymes) lower activation energy and increase reaction rates, and suggest a suitable method for investigating the effect of a given variable on reaction rate.

Key Concepts:

1. Catalyst.
2. Activation energy.
3. Activated molecules.
4. Effects of different variables on the rate of chemical rate.

Skills: Interpret data obtained from experiments concerned with rate of reaction
Essential Questions: how does the catalyst impact in our life?

Textbook and Resource Materials: how does the catalyst impact in our life?

Evidence of Learning: Suggest a suitable method for investigating the effect of a given variable on the rate of the reaction.

Capstone Connection: Catalysts and rates of reaction in water treatment

Relations:

SEC Topic & Code: HSST-CH 3.05 (HSST-CH 3.05)

CH.2.06 - Explain the meaning of reversible reaction, dynamic equilibrium, and state.

Week 11 - Week 12

Learning Outcome: Explain the meaning of reversible reaction, dynamic equilibrium, and state Le Châtelet's Principle. Apply these concepts to predict the change that will occur when a stress is applied to system at equilibrium.

Key Concepts:

1. Chemical equilibria.
2. Reversible reactions.
3. Dynamic equilibrium.
    a. factors affecting chemical equilibria.
    b. equilibrium graphing.
    c. equilibrium constants
    d. the Haber processes.
    e. the Contact processes.

Skills: 1. Illustrate and explain whether changes in concentration, pressure or temperature or the presence of a catalyst affect the value of the equilibrium constant for a reaction. 2. Deduce expressions for equilibrium constants in terms of concentrations, Kc and partial pressures, Kp 3. Calculate the values of equilibrium constants in terms of concentrations or partial pressures from appropriate data 4. Calculate the quantities present at equilibrium, given appropriate data.

Essential Questions: How does equilibrium impact in our daily life?

Textbook and Resource Materials: How does equilibrium impact in our daily life?

Evidence of Learning: - Write equilibrium law expressions from balanced chemical equations for heterogeneous and homogeneous systems. Include: mass action expression - Perform a laboratory activity to determine the equilibrium constant of an equilibrium system. - Use Le Châtelier's principle to predict and explain shifts in equilibrium. - Written quiz - individual Presentation - teamwork."

Capstone Connection: Chemical reactions in water treatment

Relations: ME.2.06; () PH.1.03; ()

SEC Topic & Code: HSST-CH 3.05. (HSST-CH 3.05.)

CH.2.07 - Explain qualitatively the extent of dissociation of strong and weak acids and.

Week 13 - Week 14

Learning Outcome: Explain qualitatively the extent of dissociation of strong and weak acids and bases. Use the common ion effect to explain buffers in terms of pH, Ka, pKa, Kw and Ksp. Calculate related quantities.

Key Concepts:

1. Ionic equilibria.
2. Brønsted-Lowry theory of acids and bases.
    a. acid dissociation constants, Ka and the use of pKa.
    b. the ionic product of water, Kw.
    c. pH: choice of pH indicators.
    d. buffer solutions.
    e. solubility product.
    f. the common ion effect.

Skills: 1. Calculate [H+(aq)] and pH values for strong and weak acids and strong bases 2. Explain the choice of suitable indicators for acid-base titrations, given appropriate data 3. Describe the changes in pH during acid-base titrations and explain these changes in terms of the strengths of the acids and bases 4. Use equations to illustrate the common ion effect 5. Explain how buffer solutions control pH and describe and explain the uses of buffers, including the role of HCO3– in controlling pH in blood 6. Calculate the pH of buffer solutions, given appropriate data 7. Explain and illustrate the use of solubility product, Ksp 8. Calculate Ksp from concentrations and vice versa

Essential Questions: How could you analyze the data of water treatment?

Textbook and Resource Materials: How could you analyze the data of water treatment?

Evidence of Learning: Describe by the equation the activation energy for the forward direction and the activation energy for the reverse direction. - Written quiz - individual Presentation - teamwork. Which is more soluble MX or MX2 where Ksp = 1.8 x 10 -6

Capstone Connection: Role of pH in potable water measurement and treatment

Relations: ME.2.07; () PH.1.03; ()

SEC Topic & Code: HSST-CH 3.05 HSST-CH 2.05 (HSST-CH 3.05 HSST-CH 2.05)